Nandish Jha

+1 (639) 998-7072 | nandish.d.jha@gmail.com | linkedin.com/in/nandish-jha | github.com/nandish-jha | Canadian Citizen

Education

Bachelor of Science in Computer Science (In Progress) — University of Saskatchewan
Specialization in Computer Systems — *Expected Apr. 2027*

Bachelor of Science in Computer Engineering — University of Saskatchewan
With Certificate in Professional Communication — *Sep. 2019 – Apr. 2024*

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, Bash, TCL, Verilog, SystemVerilog
Software & Tools: Linux, Git, Docker, React, Node.js, AWS, TensorFlow
Hardware & EDA: Cadence, Synopsys, QuestaSim, ModelSim, Quartus, Intel FPGA, Raspberry Pi, UVM, RTOS
Domains: Full-Stack Development, Embedded Systems, Functional Verification, Test Automation, Hardware-Software Co-Design

Projects

UVM-Based Functional Verification – XSwitch | *SystemVerilog, UVM, Python, TCL, QuestaSim* — 2023

- Developed complete UVM testbench achieving 95% functional coverage across 200+ test cases using constrained random stimulus and coverage-driven verification
- Automated simulation compilation, execution, and coverage reporting pipelines with Python and TCL, reducing manual execution time by 80% and enabling reproducible regression runs
- Diagnosed and reproduced 15+ critical defects with zero escapes using assertion-based debug methodology

IoT Smart System – ML-Integrated SoC Platform | *Python, Raspberry Pi 4, TensorFlow, Linux* — Jan. – Apr. 2024

- Built an IoT system with integrated machine learning for automated device control, improving system efficiency by 25% through predictive analytics
- Integrated signal acquisition, data processing, and hardware actuation subsystems with robust error handling and logging, reaching 99% uptime in deployment

Discord Lite – Full-Stack Chat Platform | *JavaScript, React, Node.js, SQL, Docker* — Jan. – Mar. 2023

- Built a full-stack chat platform with a React frontend and SQL backend, containerized with Docker for multi-service deployment
- Implemented user authentication, role-based access control, and multi-channel chat data flows; resolved environment and integration defects across the containerized stack

Low-Tech IV Pump – Capstone Design Project | *Python, MicroPython, Raspberry Pi Pico* — Sep. 2023 – Apr. 2024

- Designed a low-cost IV pump prototype for resource-constrained healthcare settings, achieving 98% fluid delivery accuracy with closed-loop control and sensor feedback
- Won the Social Impact Award at the 2024 Capstone Design Showcase

4-Bit Microprocessor – Custom MPU on FPGA | *Verilog, C, MicroC/OS-II, Intel FPGA* — 2022 – 2023

- Designed a custom 4-bit microprocessor (ALU, sequencer, instruction decoder) validated through ModelSim simulation; implemented real-time task scheduling on MicroC/OS-II RTOS

Experience

Student Senator — University of Saskatchewan Senate
Elected undergraduate representative; cross-functional policy collaboration — *Sep. 2026 – Apr. 2027*

Campus Leadership — USSU Elections Campaign Manage
Led 2024 campaign (10% turnout increase); coordinated cross-functional team of volunteers

Front End / Cash Office — Real Canadian Superstore
Cash operations, reconciliation, and customer resolution in high-volume environment — *2020 – 2021; 2024 – Present*

Certifications

AWS Certified Cloud Practitioner | UVM SystemVerilog Verification | SoC Validation & Hardware Bring-up | AI/Deep Learning (Udemy) | Claude AI Development (Anthropic)